# Analysis of PDEs

Shing Tak Lam

Michaelmas 2023*

## Contents

1 **Basics** 1
- 1.1 Examples of PDEs . . . 1
- 1.2 Data and Well-Posedness . . . 3
- 1.3 Classifying PDEs (of order $k$) . . . 3

2 **Cauchy-Kovalevskaya theorem** 4
- 2.1 ODE theory . . . 4
- 2.2 Real analyticity and majorants . . . 5

3 **Sobolev spaces** 7

4 **Elliptic PDEs** 7

5 **Hyperbolic PDEs** 7

Lecture 1

# 1 Basics

Let $\mathcal{U} \subseteq \mathbb{R}^n$ be an open set. A *PDE of order $k$* is an expression of the following form

$$F(x, u, Du, \dots, D^k u) = 0 \tag{1}$$

where $u : \mathcal{U} \to \mathbb{R}$ is the unknown, $F : \mathcal{U} \times \mathbb{R} \times \mathbb{R}^n \times \cdots \times \mathbb{R}^{n^k}$ is a general function. We say that $u$ is a *classical solution of* eq. (1) if $u$ satisfies eq. (1) in $\mathcal{U}$.

We may also consider the case where $u(x) \in \mathbb{R}^p$ and $F \in \mathbb{R}^q$, and we call this a *system of PDEs*.

## 1.1 Examples of PDEs

**Example 1.1** (ODE system)

One example of a model used in mathematical biology is the ODE system

$$\frac{\mathrm{d}u}{\mathrm{d}t} = f(u, v)$$
$$\frac{\mathrm{d}v}{\mathrm{d}t} = g(u, v)$$

**Example 1.2** (Laplace's equation)

$$\Delta u = \sum_{i=1}^{n} \partial_i^2 u = 0$$

*Based on lectures by Zoe Wyatt. Last updated October 7, 2023.

The Laplacian is an *averaging* operator.

**Example 1.3** (Heat equation)

$$u_t = D\Delta u$$

This is also called the diffusion equation, $D$ is called the diffusion constant.

**Example 1.4** (Navier-Stokes)

The Navier-Stokes equations in fluid dynamics is

$$\begin{aligned} u_t &= v\Delta u - u \cdot \nabla u - \nabla \rho + f \\ \operatorname{div}(u) &= 0 \end{aligned}$$

**Example 1.5** (Transport equation)

The transport equation is

$$u_t + v u_x = 0$$

where $v$ is a constant, corresponding to the velocity. A modification is the *advection-diffusion* equation,

$$u_t + v \cdot \nabla u = D\Delta u + f$$

**Example 1.6** (Poisson equation)

$$\Delta u = f$$

Describes electric field due to some charge, or Newtonian gravity.

**Example 1.7** (Wave equation)

$$\Box u = -u_t t + c^2 \Delta u = 0$$

This models sound waves, seismic waves, ...

**Example 1.8** (KdV equation)

This equation admits *soliton* solution.

$$u_t + \partial_x^3 u - 6u\partial_x u = 0$$

**Example 1.9** (Maxwell equations)

$$\begin{aligned}\operatorname{div}(E) &= \rho \\ \operatorname{div}(B) &= 0 \\ \partial_t E &= \nabla \times B = J \\ \partial_t B &= -\nabla \times F\end{aligned}$$

**Example 1.10** (Einstein's equations)

$$\operatorname{Ric}(g) - \frac{1}{2} g R(g) = 0$$

## 1.2 Data and Well-Posedness

All of the examples from above need additional information to solve, which we call the *data*. For example, we might need $u|_{\partial\mathcal{U}}$ and so on. A guiding principle to this process is called *well-posedness (in the sense of Hadamard)*.

We say that a PDE problem (equation and the data) is well-posed if we have

1. A solution exists (in some function space).
2. Given some data, the solution should be unique (depends on the function space of choice).
3. The solution depends continuously on the data.

The aim is to find the largest space for which a solution exists, but small enough so that it is unique. (For example, strong cosmic censorship in GR?)

**Notation 1.11** (Multi-index notation)**.** We will use multi-index notation,

$$\mathbb{N} = \{0, 1, \dots\}$$

$\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}^n$ is called a *multi-index*, and we define the *order* of $\alpha$

$$|\alpha| = \alpha_1 + \alpha_2 + \cdots + \alpha_n$$

and the $\alpha$-th derivative is

$$D^\alpha f(x) = \partial_1^{\alpha_1} \cdots \partial_n^{\alpha_n} f$$

If $x = (x_1, \dots, x_n)$, then $x^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n}$, and

$$\alpha! = \alpha_1! \cdots \alpha_n!$$

## 1.3 Classifying PDEs (of order $k$)

We say eq. (1) is *linear* if $F$ is a linear function of $x, u$ and its derivative. That is, we can write it as

$$\sum_{|\alpha| \le k} a_\alpha(x) D^\alpha u = f(x)$$

Moreover, we say that a linear PDE is *homogeneous* if $f = 0$. We say eq. (1) is *semilinear* if the highest order derivatives appear linearly with coefficients depending only on $x$. That is, we have

$$\sum_{|\alpha| = k} a_\alpha(x) D^\alpha u(x) + F(x, u, Du, \dots, D^{k-1}u) = 0$$

One example would be $\Delta u = u_x^2$. Finally, we say eq. (1) is *quasilinear* if the highest order derivatives appear linearly, but the coefficients depending on lower order derivatives. That is, we have

$$\sum_{|\alpha| = k} a_\alpha(x, u, \dots, D^{k-1}u) D^k u + F(x, u, \dots, D^{k-1}u) = 0$$

For example, we can have $uu_{xx} + u_{yy} - u_x^2 = 0$. Finally, we say eq. (1) is *fully non-linear* if it is non of the above.

# 2 Cauchy-Kovalevskaya theorem

## 2.1 ODE theory

Fix $\mathcal{U} \subseteq \mathbb{R}^n$ open, and suppose $f : \mathcal{U} \to \mathbb{R}^n$ is given. We would like to consider the ODE

$$\begin{cases} \dot{u}(t) = f(u(t)) \\ u(0) = u_0 \end{cases} \tag{2}$$

**Theorem 2.1** (Picard-Lindelöf)**.** Suppose we have $r, K > 0$ such that $B_r(u_0) \subseteq U$ and

$$\|f(x) - f(y)\| \leq K\|x - y\|$$

for all $x, y \in B_r(u_0)$. Then there exists $\varepsilon > 0$, depending on $K, r$ and a unique $C^1$ solution $u : (-\varepsilon, \varepsilon) \to \mathcal{U}$ solving eq. (2).

Lecture 2

*Sketch proof, see examples sheet 1.* If $u \in \mathbb{C}^1$ solves eq. (2), then by the fundamental theorem of calculus, $u$ satisfies the *weak formulation*

$$u(t) = u_0 + \int_0^t f(u(s))\mathrm{d}s \tag{3}$$

Moreover, if $u \in C^0$ is a solution to eq. (3), then it is a $C^1$ solution to eq. (2). Moreover, if $u$ exists, then it is a fixed point of

$$G(w) = u_0 + \int_0^t f(w(s))\mathrm{d}s$$

Let $\mathcal{S} = \left\{w : (-\varepsilon, \varepsilon) \to \overline{B_{r/2}(u_0)}\right\}$ continuous. We want to show that $\mathcal{S}$ is a complete metric space, $G : \mathcal{S} \to \mathcal{S}$ is a contraction for $\varepsilon$ sufficiently small, and by the contraction mapping theorem $G$ has a fixed point. □

**Remark 2.2.**

1. The solution (in general) can't be global. Consider for example
$$\dot{u}(t) = u(t)^2 \quad \text{with} \quad u(0) = u_0 > 0$$
Solutions to this equation blow up in finite time.
2. This does not apply to
$$\dot{u}(t) = \sqrt{u(t)} \quad \text{with} \quad u(0) = 0$$
There are two solutions. Note we *can* apply the Peano existence theorem.

Now suppose $f$ is smooth, and we have $\dot{u}(t) = f(u(t))$ is $C^1$. By the chain rule,

$$\ddot{u}(t) = Df(u(t)) \cdot \dot{u}(t) = f_2(u(t), \dot{u}(t))$$

which is continuous. Hence $\ddot{u}$ is continuous, and so $u \in C^2$. Repeating this, we get that $u \in C^k$ for all $k$. That is, $u$ is smooth.

In principle, given $u_0 = u(0)$, we can determine

$$u^{(k)}(0) = F_k\left(u, u', \ldots, u^{(k-1)}\right)\Big|_{t=0}$$

and so we can write

$$\sum_{k \geq 0} \frac{u^{(k)}(0)}{k!} t^k$$

We call this a *formal power series solution*. Does our solution $u(t)$ agree with this? That is, do we have

$$u(t) = \sum_{k\geq 0} \frac{u^{(k)}(0)}{k!} t^k$$

in a neighbourhood of 0?

**Theorem 2.3** (Cauchy-Kovalevskaya for simple ODEs). If $f(u)$ is real analytic in a neighbourhood of $u_0$, then the series

$$\sum_{k\geq 0} \frac{u^{(k)}(0)}{k!} t^k$$

converges in a neighbourhood of 0 to the unique solution of eq. (2) given by Picard-Lindelöf.

## 2.2 Real analyticity and majorants

Suppose $f : (-\varepsilon, \varepsilon) \to \mathbb{R}$ is a smooth function. Therefore, $f^{(n)}(0)$ exists for all $n \geq 0$. Does the partial sums

$$\sum_{n\geq 0} \left|f^{(n)}(0)\right| n! x^n$$

for some $|x| \leq \delta$? No, consider the function

$$f(x) = \begin{cases} \exp\left(-\frac{1}{x}\right) & x > 0 \\ 0 & x \leq 0 \end{cases}$$

This is a smooth function, with $f^{(n)}(0) = 0$ for all $n$.

**Definition 2.4** (real analytic)
Let $\mathcal{U} \subseteq \mathbb{R}^n$ be open and $f : \mathcal{U} \to \mathbb{R}$ is *real analytic* (at $x_0$) if there exists $r > 0$, $f_\alpha \in \mathbb{R}$ such that

$$\sum_{\alpha} f_\alpha (x - x_0)^\alpha$$

for all $|x - x_0| < r$.

**Remark 2.5.**

1. That is, $f$ can be written as a convergent power series and
$$f_\alpha = \frac{D^\alpha f(x_0)}{n!}$$
2. If $f$ is real analytic about $x_0 \in \mathcal{U}$, then $f$ is real analytic about all of $\mathcal{U}$.
3. We will denote the set of real analytic functions on $\mathcal{U}$ by $C^\omega(\mathcal{U})$.
4. If $f$ is $C^\omega$, then $f$ is smooth (e.g. Weierstrass $M$-test).
5. If $f$ is real analytic, and $\mathcal{U}$ is connected, then $f$ is uniquely determined in $\mathcal{U}$ by its derivatives $D^\alpha f(x)$ at some point $x \in \mathcal{U}$.

Exercise: Show $f(x) = 1/x$ and $f(x) = \sqrt{x}$ are real analytic for $x > 0$.

**Example 2.6**
Recall

$$\frac{1}{1-x} = \sum_{k\geq 0} x^k$$

for $|x| < 1$. Let $r > 0$, and consider

$$f(x) = \frac{r}{r - (x_1 + \cdots + x_n)} = \frac{1}{1 - \frac{x_1 + \cdots + x_n}{r}} = \sum_{k \geq 0} \left( \frac{x_1 + \cdots + x_n}{r} \right)^k$$

provided $|x_1 + \cdots + x_n| \leq \sqrt{n} \left( \sum_j |x_j|^2 \right)^{1/2} = \sqrt{n} ||x|| < r$. By the multinomial theorem (sheet 1),

$$f(x) = \sum_{k \geq 0} \frac{1}{r^k} \sum_{|\alpha| = k} \binom{|\alpha|}{\alpha} x^\alpha = \sum_\alpha \frac{|\alpha|!}{\alpha! r^{|\alpha|}} x^\alpha$$

where

$$\binom{|\alpha|}{\alpha} = \frac{|\alpha|!}{\alpha!}$$

and so,

$$f(x) = \sum_\alpha f_\alpha x^\alpha$$

where

$$f_\alpha = \frac{|\alpha|!}{\alpha!} \frac{1}{r^{|\alpha|}}$$

This series is absolutely convergent near zero, since

$$\sum_\alpha \frac{|\alpha!|}{\alpha!} \frac{x^\alpha}{r^{|\alpha|}} = \sum_{k \geq 0} \left( \frac{|x_1| + \cdots + |x_n|}{r} \right)^k < \infty$$

**Definition 2.7** (majorise)
Let $f = \sum_\alpha f_\alpha x^\alpha$ and $g = \sum_\alpha g_\alpha x^\alpha$, we say that $g$ *majorises* $f$, or $g$ *is a mojorant of* $f$, written $g \gg f$, if $g_\alpha \geq |f_\alpha|$ for all $\alpha$.

For vector valued functions, we require each component to be a majorant.

**Lemma 2.8** (properties of majorants).
1. If $g \gg f$, and $g$ converges for $||x|| < r$, then $f$ converges for $||x|| < r$.
2. If $f = \sum f_\alpha x^\alpha$ converges for $||x|| < r$, then for any $s \in (0, r/\sqrt{n})$, there exists a majorant of $f$ which converges for $||x|| < s/\sqrt{n}$.

*Proof.* 1. Looking at the partial sums

$$\sum_{|\alpha| \leq k} |f_\alpha x^\alpha| = \sum_{|\alpha| \leq k} |f_\alpha| |x_1|^{\alpha_1} \cdots |x_n|^{\alpha_n} \leq \sum_{|\alpha| \leq k} |g_\alpha| |x_1|^{\alpha_1} \cdots |x_n|^{\alpha_n} \leq \sum_\alpha |x_1|^{\alpha_1} \cdots |x_n|^{\alpha_n} = g(\tilde{x})$$

where $\tilde{(x)} = (|x_1|, \ldots, |x_n|)$. So $\left\| \tilde{x} \right\| = ||x||$, and so if $||x|| < r$, then $g$ converges at $\tilde{x}$. That is, $g(\tilde{x}) < \infty$. Therefore, we have a uniform bound on the partial sum.

2. Let $s \in (0, r/\sqrt{n})$, and set $y = (s, \ldots, s)$. Then $\left\| y \right\| = s\sqrt{n}$, and by assumption,

$$f(y) = \sum_\alpha f_\alpha y^\alpha$$

converges, as $\left\| y \right\| = s\sqrt{n} < r$. So there exists a constant $c$ such that $|f_\alpha y^\alpha| \leq c$. Hence

$$|f_\alpha| \leq \frac{C}{|y^\alpha|} = \frac{C}{|y_1|^{\alpha_1} \cdots |y_n|^{\alpha_n}} = \frac{C}{|s|^{|\alpha|}} \leq \frac{C}{|s|^{|\alpha|}} \frac{|\alpha|!}{\alpha!}$$

So we define

$$g(x) = \frac{Cs}{s - (x_1 + \cdots + x_n)}$$

From the above, $g$ majorises $f$. □

3 Sobolev spaces

4 Elliptic PDEs

5 Hyperbolic PDEs

# Index

Cauchy-Kovalevskaya Theorem
    for simple ODEs, 5
classical solution, 1

data, 3

fully non-linear PDE, 4

linear PDE, 3
    homogeneous, 3

majorise, 6
multi-index notation, 3

PDE of order $k$, 1
Picard-Lindelöf, 4

quasilinear PDE, 3

real analytic, 5

semilinear PDE, 3
system of PDEs, 1

well-posed, 3